AgriYard - Flutter Project Presentation Guide

Complete Widget Architecture, Purpose & Code File Mapping for College Viva

Key Presentation Tip for College Professor:
*Explain that the entire project follows Flutter's **Declarative Widget Tree Architecture**. Every screen starts with a Scaffold, structured vertically with Column or scrollable lists (ListView.builder / GridView.builder), and styled with reusable custom components (CustomPrimaryButton, CustomInputField, AppCurvedHeader).*

Part 1: Flutter Widgets Dictionary (Meaning & Exact Purpose)

When the professor asks: *'What is this widget and why did you use it?'*, refer to the answers below:

Widget Name	What it is (Role)	Exact Purpose in AgriYard App
Scaffold	Base layout structure of a screen	Provides background color, AppBar slot, Body area, and BottomNavigationBar slot.
SafeArea	Invasive UI padding shield	Prevents content from getting clipped by mobile camera notches and top status bars.
Column	Vertical layout container	Arranges child widgets in a vertical line (top-to-bottom), like Logo above input fields.
Row	Horizontal layout container	Arranges child widgets side-by-side, e.g., Search bar next to Date Picker, or Min/Max prices.
SingleChildScrollView	Scrollable single-child wrapper	Allows scrolling when screen content overflows or when mobile keyboard opens.
ListView.builder	Dynamic lazy-loading list	Efficiently renders large lists (Crop Rates, Shops, Users) only loading visible items.
GridView.builder	2D grid layout generator	Creates the 2x2 grid cards of Marketing Yards (Rajkot, Junagadh, Gondal, Mendarda) on Home.
Container	Universal box styling model	Sets custom width, height, padding, borders, background colors, and BoxShadows.
Expanded	Flex space consumer	Expands a child of Row/Column to fill all remaining available space evenly.
Stack / Positioned	Overlapping layer system	Used on Onboarding (Farmer & Tractor placed on green hill) and Home cards (gradient over images).
Card	Material surface container	Displays distinct content sections with rounded corners and elevation/shadow.
TextField / TextFormField	User text input field	Accepts inputs with validation in Login, Signup, Forgot Password, and Shop Search.
ElevatedButton	Primary raised action button	Main solid action buttons (Login, Create Account, Apply Filters, Send Reset Link).

OutlinedButton	Bordered secondary button	Used for secondary actions like Logout button and secondary screen options.
TextButton	Border-free clickable text	Used for text-only links like 'Forgot Password?' and 'Sign Up' prompts.
ListTile	Standardized list item row	Displays leading icon, title, subtitle, and trailing arrow in Profile and Settings menus.
DataTable / PriceTable	Tabular multi-column grid	Renders 7-day historical prices (Date, Min, Avg, Max) on Crop Detail screens.
CircularProgressIndicator	Activity spinner loader	Shows animated spinning circle on Splash Screen while app prepares data.

Part 2: Screen-by-Screen Code File Mapping

Exact file paths, widgets used, and location references for all screens:

1. Splash / Loading Screen

File: lib/screens/auth/splash_screen.dart
Key Widgets: Scaffold, Column, FadeTransition, ScaleTransition, Image.asset, CircularProgressIndicator
Purpose: Initial launch screen with animated AgriYard logo, tagline, and loading spinner.

2. Welcome / Onboarding Screen

File: lib/screens/auth/onboarding_screen.dart
Key Widgets: Scaffold, SafeArea, SingleChildScrollView, Stack, Container, Image.asset, CustomPrimaryButton
Purpose: Welcomes user with farmer/tractor illustration, 3 core feature highlights, and 'Get Started' action.

3. Login Screen

File: lib/screens/auth/login_screen.dart
Key Widgets: Scaffold, Form, Column, CustomInputField, TextButton (Forgot Password), CustomPrimaryButton
Purpose: User authentication screen with email & password validation and navigation to Signup & Forgot Password.

4. Forgot Password Screen

File: lib/screens/auth/forgot_password_screen.dart
Key Widgets: Scaffold, AppBar, Container (Lock icon), CustomInputField, CustomPrimaryButton
Purpose: Allows users to enter their email/mobile to receive password reset link.

5. Register / Create Account Screen

File: lib/screens/auth/signup_screen.dart
Key Widgets: Scaffold, Form, SingleChildScrollView, 5x CustomInputField, CustomPrimaryButton
Purpose: Account registration with fields for Full Name, Mobile, Email, Password, and Confirm Password.

6. Main Dashboard / Home Screen

File: lib/screens/home/home_screen.dart
Key Widgets: AppCurvedHeader, Row, Search Container, GridView.builder (2x2 Yards), Price Rise & Fall Action Cards
Purpose: Core dashboard featuring curved green header, quick search, date selector, 2x2 marketing yards grid, and trend cards.

7. Search Results Screen

File: lib/screens/home/search_results_screen.dart
Key Widgets: AppBar (with search TextField), ListView.separated, Result Card Container, Image.asset
Purpose: Live search results displaying matching crops across yards with rates and quick navigation.

8. Filter & Sort Screen

File: lib/screens/home/filter_sort_screen.dart
Key Widgets: AppBar, Custom Radio/Circle (Sort), Custom Checkboxes (Yards), CustomPrimaryButton
Purpose: Allows sorting by price (high/low) and filtering by specific yards (Rajkot, Gondal, Junagadh, Mendarda).

9. Marketing Yard Details (Rajkot / Gondal / Mendarda)

File: lib/screens/yards/yard_detail_screen.dart
Key Widgets: AppBar, Image.asset (Yard photo), 4x Info Rows (Address, Phone, Shops, Est.), Row of 2 Action Buttons
Purpose: Full profile of selected marketing yard with direct buttons to view Today's Rates and Shop List.

10. Today's Market Rates Screen

File: lib/screens/market/market_rates_screen.dart
Key Widgets: AppBar, ListView.separated, Crop Card Container, Row (Arrival, Min, Avg, Max), ElevatedButton
Purpose: Comprehensive list of all crops with daily arrival volume, minimum, average, and maximum prices.

11. Crop Detail Screen (Wheat, Soyabean, Chana, Tuver, Garlic)

File: lib/screens/market/crop_detail_screen.dart
Key Widgets: Image Container, Summary Badge (20kg), 2x Metric Rows (Total arrival, Min/Avg/Max), PriceTable
Purpose: Detailed crop analysis page showing 20kg pricing, arrival stats, price capsule, and 7-day historical rate table.

12. Price Rise / Top Gainers Screen

File: lib/screens/market/price_rise_screen.dart
Key Widgets: AppBar, Header Icon, ListView.separated, Green Gain Badge Container, Price Stats Row
Purpose: Displays crops with highest upward price momentum today with percentage increase badges.

13. Shop List Screen

File: lib/screens/shops/shop_list_screen.dart
Key Widgets: AppBar, TextField (Search), ListView.separated, Shop Card Container, IconButton (Direct Phone Call)
Purpose: Directory of trusted APMC market yard shops with shop numbers, merchant names, and one-tap calling.

14. User Profile Screen

File: lib/screens/profile/profile_screen.dart
Key Widgets: Curved Header, CircleAvatar, Text, ListTile menu (Personal Info, Password, Help, Logout)
Purpose: User account settings with profile picture, personal details, and links to change password and support.

15. Change Password Screen

File: lib/screens/profile/change_password_screen.dart
Key Widgets: AppBar, 3x CustomInputField (Current, New, Confirm), CustomPrimaryButton
Purpose: Form allowing users to update their login credentials securely.

16. Admin Dashboard Screen

File: lib/screens/admin/admin_dashboard_screen.dart
Key Widgets: AppBar, Stat Cards (Users, Active Yards), DropdownButtonFormField, InputFields (Quick Rate Update)
Purpose: Admin control center for quick price updates, viewing system metrics, and selecting yards.

17. Admin Manage Data Screen

File: lib/screens/admin/admin_manage_data_screen.dart
Key Widgets: TabBar (Yards, Crops, Shops), ElevatedButton (Add New), ListView.builder, Edit/Delete IconButtons
Purpose: CRUD management screen for creating, editing, and deleting yards, crops, and shops.

18. Admin Users & Alerts Screen

File: lib/screens/admin/admin_users_alerts_screen.dart
Key Widgets: TextField (Push Message), ElevatedButton (Send), ListView.builder, Status Chip (Active/Blocked)
Purpose: Broadcasting push notifications and managing registered user accounts with active/blocked toggle.

19. Admin Settings Screen

File: lib/screens/admin/admin_settings_screen.dart
Key Widgets: AppBar, ListTiles (Admin Password, Helpline, Terms, Version), OutlinedButton (Logout)
Purpose: Administrative configuration screen for system credentials, contact info, and app settings.

20. Main Navigation Wrapper

File: lib/screens/main_navigation_screen.dart
Key Widgets: Scaffold, IndexedStack (Preserves Tab States), CustomBottomNav (5 Persistent Tabs)
Purpose: Root navigation shell managing the 5 primary tabs: Home, Market Rates, Price Rise, Shops, Profile.